

IGCSE Chemistry

Comprehensive Study Notes

Chemical Reactions · Atomic Structure · Periodic Table

Topics Covered	Level	Edition
3 Core Chapters	IGCSE / O-Level	2024 – 2026

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Table of Contents

Chapter 1 Introduction to Chemical Reactions

- 1.1 What is a Chemical Reaction?
- 1.2 Signs of a Chemical Reaction
- 1.3 Types of Chemical Reactions
- 1.4 Word & Symbol Equations
- 1.5 Balancing Equations
- 1.6 Examinable Examples & Worked Questions
- 1.7 Practice Exercises

Chapter 2 Atomic Structure

- 2.1 Sub-atomic Particles
- 2.2 Atomic Number & Mass Number
- 2.3 Isotopes
- 2.4 Electronic Configuration
- 2.5 Ions & Ionic Charge
- 2.6 Examinable Examples & Worked Questions
- 2.7 Practice Exercises

Chapter 3 The Periodic Table

- 3.1 Arrangement of Elements
- 3.2 Periods and Groups
- 3.3 Group 1 – Alkali Metals
- 3.4 Group 7 – Halogens
- 3.5 Group 0 – Noble Gases
- 3.6 Transition Metals
- 3.7 Trends Across a Period
- 3.8 Examinable Examples & Worked Questions
- 3.9 Practice Exercises

Chapter 1

Introduction to Chemical Reactions

Chemical reactions are fundamental to all of chemistry. In this chapter you will learn to identify, classify, write and balance chemical equations – core skills tested heavily in IGCSE Paper 2 (Theory) and Paper 4 (Extended).

1.1 What is a Chemical Reaction?

A **chemical reaction** is a process in which substances called **reactants** are converted into new substances called **products**. Atoms are *rearranged* but never created or destroyed (Law of Conservation of Mass). The total mass of reactants always equals the total mass of products.

■ Definition: Chemical Reaction

A process where reactants are chemically changed into products with different chemical properties. Bonds in reactants are broken and new bonds in products are formed.

1.2 Signs of a Chemical Reaction

Sign	Example
Colour change	Iron rusting turns orange-brown; copper sulfate solution turns blue when iron is added.
Gas produced	Bubbles of CO_2 when acid reacts with a carbonate.
Precipitate formed	A white precipitate of BaSO_4 forms when Ba^{2+} and SO_4^{2-} ions meet.
Temperature change	Combustion releases heat (exothermic); dissolving NH_4Cl absorbs heat (endothermic).
Light emitted	Magnesium burns with a bright white flame.
Irreversibility	Cooking an egg — a physical change can be reversed; most chemical changes cannot.

1.3 Types of Chemical Reactions

Type	General Form	Example Equation	Key Feature
Synthesis (Combination)	$\text{A} + \text{B} \rightarrow \text{AB}$	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	Two or more substances combine to form one product.
Decomposition	$\text{AB} \rightarrow \text{A} + \text{B}$	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$	One compound breaks down into simpler substances.

Combustion	Fuel + O ₂ → CO ₂ + H ₂ O	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$	Rapid reaction with oxygen, releasing energy.
Neutralisation	Acid + Base → Salt + Water	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	Acid and alkali react to form a salt and water.
Oxidation / Reduction	Involves electron transfer	$\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$	Redox reactions – vital for electrochemistry topics.
Precipitation	Two solutions → insoluble solid	$\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$	A precipitate forms from mixing two aqueous solutions.

1.4 Word and Symbol Equations

Word equations name the reactants and products. **Symbol equations** use chemical formulae and must be *balanced*. State symbols are required at Extended level: (s) solid, (l) liquid, (g) gas, (aq) aqueous.

Worked Example 1.1 – Writing a Symbol Equation

Word equation: magnesium + hydrochloric acid → magnesium chloride + hydrogen

Unbalanced: $\text{Mg} + \text{HCl} \rightarrow \text{MgCl} + \text{H}$

Balanced: $\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$

Check: Mg=1✓, H=2✓, Cl=2✓ — Atom counts balance on both sides.

1.5 Balancing Equations – Step-by-Step Method

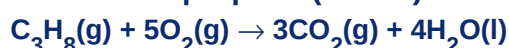
- Step 1: Write the correct formulae for all reactants and products.
- Step 2: Count atoms of each element on both sides.
- Step 3: Add coefficients (whole numbers in front of formulae) to balance — NEVER change the subscripts.
- Step 4: Recount to verify all atoms balance.
- Step 5: Add state symbols if required.

Examiner's Tip

IGCSE examiners often award 1 mark for correct formulae and 1 mark for balanced coefficients. Write correct formulae first — you gain marks even if you cannot complete the balance.

1.6 Examinable Examples & Worked Questions

Combustion of propane (C₃H₈)



Thermal decomposition of calcium carbonate



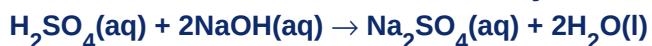
Reaction of zinc with dilute sulfuric acid



Formation of iron(III) oxide (rusting simplified)

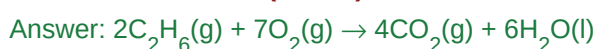


Neutralisation: sulfuric acid + sodium hydroxide



Past-Paper Style Questions

Q1 [2 marks] Write a balanced symbol equation, including state symbols, for the complete combustion of ethane (C₂H₆).



Q2 [2 marks] Iron reacts with steam to form iron(II,III) oxide and hydrogen. Balance the equation: Fe + H₂O → Fe₃O₄ + H₂



Q3 [1 mark] State two observations that indicate a chemical reaction has occurred when zinc is added to copper(II) sulfate solution.

Answer: ① Blue colour of solution fades / becomes colourless. ② Pink/orange-brown solid (copper) coats the zinc / deposits appear.

1.7 Practice Exercises

Exercise 1.1: Balance: Al + O₂ → Al₂O₃

Exercise 1.2: Balance and add state symbols: Na + H₂O → NaOH + H₂

Exercise 1.3: Write the word equation and balanced symbol equation for the reaction of calcium with water to form calcium hydroxide and hydrogen gas.

Exercise 1.4: Identify the type of reaction: Fe₂O₃ + 3CO → 2Fe + 3CO₂. Explain your answer.

Exercise 1.5: A student mixes lead(II) nitrate solution with potassium iodide solution. Predict what they would observe and write a balanced ionic equation.

Common Mistake

Never change subscripts when balancing — always change coefficients. Writing H₂ instead of H₂O is wrong chemistry!

Chapter 2

Atomic Structure

Understanding atomic structure is the foundation of all chemistry. This chapter covers sub-atomic particles, electronic configuration, isotopes and ions — all essential for Paper 1 (MCQ) and Paper 2 (Theory).

2.1 Sub-atomic Particles

All atoms consist of a central **nucleus** (containing protons and neutrons) surrounded by **electrons** arranged in shells (energy levels). The atom is mostly empty space — the nucleus is ~10,000 times smaller than the atom.

Particle	Symbol	Relative Mass	Relative Charge	Location
Proton	p	1	+1	Nucleus
Neutron	n	1	0 (neutral)	Nucleus
Electron	e ⁻	1/1840 (≈0)	-1	Shells around nucleus

2.2 Atomic Number and Mass Number

■ Definition: Atomic Number (Z)

The number of PROTONS in the nucleus. Defines the element. In a neutral atom, number of protons = number of electrons.

■ Definition: Mass Number (A)

The total number of protons + neutrons in the nucleus. Number of neutrons = $A - Z$.

Worked Example 2.1 – Deducing Sub-atomic Particles

Element: Chlorine-35 Notation: $^{35}_{17}\text{Cl}$

Atomic number $Z = 17 \rightarrow$ Protons = 17

Mass number $A = 35 \rightarrow$ Neutrons = $35 - 17 = 18$

Neutral atom $\rightarrow$ Electrons = 17

For the Cl^{-} ion: $\rightarrow$ Electrons = $17 + 1 = 18$ (one extra electron gained)

2.3 Isotopes

■ Definition: Isotopes

Atoms of the SAME element with the SAME number of protons but DIFFERENT numbers of neutrons (and therefore different mass numbers).

Isotopes have **identical chemical properties** (same electron configuration) but **different physical properties** such as mass, density, and rate of diffusion.

Isotope	Protons	Neutrons	Electrons	Mass Number
Carbon-12 (^{12}C)	6	6	6	12
Carbon-13 (^{13}C)	6	7	6	13
Carbon-14 (^{14}C)	6	8	6	14
Chlorine-35 (^{35}Cl)	17	18	17	35
Chlorine-37 (^{37}Cl)	17	20	17	37

■ Examiner's Tip

Relative atomic mass (A_r) is the weighted mean mass of an element's isotopes relative to 1/12 of carbon-12. For Cl: $A_r = (35 \times 75/100) + (37 \times 25/100) = 35.5$. You may be asked to calculate this from percentage abundances.

2.4 Electronic Configuration

Electrons occupy **shells (energy levels)** around the nucleus. Each shell holds a maximum number of electrons:

- Shell 1 (innermost): maximum 2 electrons
- Shell 2: maximum 8 electrons
- Shell 3: maximum 8 electrons (at IGCSE level)

Element	Symbol	Z	Electronic Config.	Diagram Notation
Hydrogen	H	1	1	1
Helium	He	2	2	2
Lithium	Li	3	2, 1	2.1
Carbon	C	6	2, 4	2.4
Nitrogen	N	7	2, 5	2.5
Oxygen	O	8	2, 6	2.6
Sodium	Na	11	2, 8, 1	2.8.1
Magnesium	Mg	12	2, 8, 2	2.8.2
Chlorine	Cl	17	2, 8, 7	2.8.7
Argon	Ar	18	2, 8, 8	2.8.8

Calcium	Ca	20	2, 8, 8, 2	2.8.8.2
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2.5 Ions and Ionic Charge

Atoms become **ions** by gaining or losing electrons. Metal atoms *lose* electrons to form positive cations; non-metal atoms *gain* electrons to form negative anions.

Ion	Protons	Electrons	Charge	Config
Na ⁺	11	10	+1	2.8
Mg ²⁺	12	10	+2	2.8
Al ³⁺	13	10	+3	2.8
Cl ⁻	17	18	-1	2.8.8
O ²⁻	8	10	-2	2.8
N ³⁻	7	10	-3	2.8

2.6 Examinable Examples & Worked Questions

Q1 [3 marks] An atom of element X has the electronic configuration 2, 8, 6. State the number of protons, the period and the group of element X.

Answer: Protons = 16 (total electrons in neutral atom: 2+8+6). Period = 3 (3 shells occupied). Group = VI (6 electrons in outermost shell).

Q2 [2 marks] Explain why ³⁵Cl and ³⁷Cl are isotopes of chlorine.

Answer: Both have 17 protons (same atomic number / same element) but different numbers of neutrons (18 and 20 respectively), giving different mass numbers.

Q3 [2 marks] An element has two isotopes: 60% of mass 20 and 40% of mass 22. Calculate the relative atomic mass.

Answer: $A_r = (20 \times 0.60) + (22 \times 0.40) = 12.0 + 8.8 = 20.8$

2.7 Practice Exercises

Exercise 2.1: An atom has 15 protons and 16 neutrons. State its: (a) atomic number, (b) mass number, (c) electronic configuration.

Exercise 2.2: Draw the electronic structure of a phosphorus atom (Z = 15).

Exercise 2.3: The element bromine has two isotopes: ⁷⁹Br (50.7%) and ⁸¹Br (49.3%). Calculate the A_r of bromine to 1 d.p.

Exercise 2.4: Write the symbol for the ion formed when sulfur (Z=16) gains 2 electrons, and give its electronic configuration.

Exercise 2.5: Explain why isotopes of the same element have identical chemical properties.

Chapter 3

The Periodic Table

The Periodic Table is one of chemistry's greatest organisational tools. In IGCSE, you need to know how elements are arranged, the key properties of Groups 1, 7, and 0, transition metals, and trends across periods.

3.1 Arrangement of Elements

Elements are arranged in order of increasing **atomic number (proton number)**. This arrangement places elements with similar chemical properties in the same vertical column (**group**) and reveals trends across horizontal rows (**periods**).

■ Definition: Period

A horizontal row in the periodic table. All elements in the same period have the same number of electron shells. Period number = number of shells.

■ Definition: Group

A vertical column in the periodic table. Elements in the same group have the same number of outer-shell electrons and thus similar chemical properties. Group number (1–7) = number of outer electrons (Group 0 = 8 for noble gases, 2 for He).

3.2 Periods and Groups – Key Facts

Feature	Periods	Groups
Direction	Horizontal (left to right)	Vertical (top to bottom)
What increases?	Atomic number; nuclear charge	Atomic number; number of shells
Elements share?	Same number of electron shells	Same number of outer electrons
Numbered?	1 to 7	I to VII (and 0 for noble gases)
Example	Period 3: Na, Mg, Al, Si, P, S, Cl, Ar	Group I: Li, Na, K, Rb, Cs

3.3 Group 1 – Alkali Metals

Group 1 elements (Li, Na, K, Rb, Cs) have **1 outer electron**, which they readily lose to form 1+ ions. They are stored under oil to prevent reaction with air/water.

- Soft metals, low density (Li, Na, K float on water).
- React vigorously with water → metal hydroxide + hydrogen gas.
- React with oxygen → metal oxide (Li $\blacksquare$ O); Na burns to form Na $\blacksquare$ O $\blacksquare$.

- React with halogens → metal halide salt.
- React with water more vigorously down the group.

Property	Li → Cs Trend	Reason
Reactivity	Increases	Outer electron further from nucleus, less attracted, lost more easily
Melting point	Decreases	Weaker metallic bonding as atoms get larger
Density	Generally increases	Larger, heavier atoms
Atomic radius	Increases	More electron shells added

Worked Example 3.1 – Reaction of Potassium with Water

Word equation: potassium + water → potassium hydroxide + hydrogen

Symbol equation: $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$

Observations: vigorous fizzing; K melts into silver ball; purple/lilac flame; K moves rapidly on the surface; solution turns alkaline (red litmus → blue).

3.4 Group 7 – Halogens

Halogens (F, Cl, Br, I, At) have **7 outer electrons** and readily gain one electron to form X^- ions. They exist as diatomic molecules (X_2).

Halogen	Formula	State (r.t.p.)	Colour	Reactivity
Fluorine	F_2	Gas	Pale yellow	Most reactive
Chlorine	Cl_2	Gas	Yellow-green	Very reactive
Bromine	Br_2	Liquid	Red-brown	Moderately reactive
Iodine	I_2	Solid	Grey/violet vapour	Less reactive

Displacement Reactions of Halogens

A more reactive halogen displaces a less reactive halide from solution.

$\text{Cl}_2 + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2$ — orange/brown colour appears

$\text{Cl}_2 + 2\text{KI(aq)} \rightarrow 2\text{KCl(aq)} + \text{I}_2$ — brown/violet colour appears

$\text{Br}_2 + \text{KI(aq)} \rightarrow \text{KBr(aq)} + \text{I}_2$ — brown colour appears

Br_2 CANNOT displace Cl_2 (Cl_2 is more reactive than Br_2)

Examiner's Tip

To test for halide ions use silver nitrate solution (AgNO_3) + dilute HNO_3 : $\text{Cl}^- \rightarrow$ white ppt (AgCl); $\text{Br}^- \rightarrow$ cream ppt (AgBr); $\text{I}^- \rightarrow$ yellow ppt (AgI). These are favourite exam questions!

3.5 Group 0 – Noble Gases

Noble gases (He, Ne, Ar, Kr, Xe, Rn) have **full outer shells** (2 for He; 8 for all others) making them extremely stable and **chemically inert** under normal conditions.

- Exist as monatomic gases (single atoms, not molecules).
- Used in filament lamps (Ar), balloons (He), advertising signs (Ne), lasers (Ar, Kr).
- Do not form ions or compounds under standard conditions.
- Boiling points increase down the group (larger atoms, stronger van der Waals' forces).

3.6 Transition Metals

Transition metals occupy the d-block between Groups 2 and 3. They share distinctive properties that differ from Group 1 metals.

- High melting and boiling points; high density; hard.
- Good conductors of heat and electricity.
- Form coloured compounds (e.g. CuSO_4 — blue; FeCl_3 — yellow-brown; $\text{K}_2\text{Cr}_2\text{O}_7$ — orange).
- Can form ions with different charges (variable valency): Fe^{2+} and Fe^{3+} ; Cu^+ and Cu^{2+} .
- Act as catalysts: Fe in Haber process; Pt/Rh in catalytic converters; MnO_2 in H_2O_2 decomposition.

3.7 Trends Across Period 3 (Na → Ar)

Property	Trend across Period 3	Explanation
Atomic radius	Decreases	More protons → stronger nuclear pull on electrons
Ionisation energy	Generally increases	Stronger attraction between nucleus and outer electron
Electronegativity	Increases	Atoms pull bonding electrons more strongly
Metallic character	Decreases (Na→Mg→Al are metals; Si is a metalloid; P, S, Cl, Ar are non-metals)	Size increases, so it is harder to lose electrons
Melting point	Complex trend (peaks at Si)	Changes in structure from metallic → giant covalent → simple molecules
Oxides	Basic (Na_2O) → Amphoteric (Al_2O_3) → Acidic (SO_2 , SO_3 , Cl_2O_7)	Non-metal oxides are acidic; metal oxides are basic

3.8 Examinable Examples & Worked Questions

Q1 [3 marks] Chlorine water is added to separate solutions of potassium bromide and potassium iodide. Describe what you would observe in each case and write the ionic equation for each reaction.

Answer: KBr: orange/brown colour — $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$. KI: brown/violet colour — $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$. Chlorine is more reactive and displaces both bromide and iodide ions.

Q2 [4 marks] Describe and explain the trend in reactivity of Group 1 metals from lithium to caesium.

Answer: Reactivity increases from Li to Cs. Going down the group, the number of electron shells increases, so the outer electron is further from the nucleus and is more shielded by inner shells. The nuclear attraction on the outer electron decreases, so it is lost more easily, making the metal more reactive.

Q3 [2 marks] Give two properties of transition metals that differ from Group 1 metals.

Answer: Any two of: higher melting point / higher density / harder / variable valency / form coloured compounds / act as catalysts.

Q4 [2 marks] Element Q is in Period 3, Group VI. State the number of protons and the electronic configuration of Q.

Answer: Period 3, Group VI → 3 shells, 6 outer electrons. Electronic configuration: 2.8.6 → Protons = 16 (sulfur, S).

3.9 Practice Exercises

Exercise 3.1: Potassium reacts with water. Write the word equation, balanced symbol equation, and list three observations.

Exercise 3.2: Explain why the noble gases are unreactive. Why is helium used in balloons rather than hydrogen?

Exercise 3.3: Predict whether bromine water will react with (a) potassium chloride solution and (b) potassium iodide solution. Write ionic equations for any reactions that occur.

Exercise 3.4: Transition metal compounds are often coloured. Give the colour of: (a) copper(II) sulfate solution, (b) iron(III) chloride solution, (c) potassium dichromate(VI) solution.

Exercise 3.5: An element has the following properties: high melting point, forms a +2 and +4 ion, forms coloured compounds, and is used as a catalyst. Identify the most likely group this element belongs to and give one example element.

Exercise 3.6: Complete the table: Period 3 element with 14 protons — name, symbol, group, period, and electronic configuration.

Quick-Revision Summary

Chemical Reactions

- ✓ Law of conservation of mass: total mass of reactants = total mass of products.
- ✓ Types: synthesis, decomposition, combustion, neutralisation, oxidation/reduction, precipitation.
- ✓ Balancing: change coefficients only; verify atom counts; add state symbols (s), (l), (g), (aq).

Atomic Structure

- ✓ Proton = +1, mass 1; Neutron = 0, mass 1 — both in nucleus. Electron = -1, negligible mass.
- ✓ Atomic number Z = protons; Mass number A = protons + neutrons; Neutrons = $A - Z$.
- ✓ Isotopes: same Z , different A , same chemical properties.
- ✓ Shells: max 2, 8, 8, ... electrons. Group number = outer electrons; Period = shells.

Periodic Table

- ✓ Elements ordered by increasing atomic number.
- ✓ Group 1: reactivity increases ↓ (outer e⁻ further from nucleus, easier to lose).
- ✓ Group 7: reactivity decreases ↓ (harder to attract an extra electron to larger atom).
- ✓ Noble gases: full outer shells → inert.
- ✓ Transition metals: high MP, variable valency, coloured compounds, catalysts.

■ Examiner's Tip

In every IGCSE Chemistry exam, READ THE QUESTION CAREFULLY. Use the command word: 'state' = brief fact; 'explain' = reason required; 'describe' = observations; 'calculate' = show working and units. Each mark corresponds to one distinct point — never waffle!